MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 719A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Clipper diode, high vacuum

See figure 1

Mounting position: Vertical, base up or down

Weight: 6-ounces (170.1 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Pp	tk	Pulsing service
Unit:	V	kv	a	W	sec	---
Maximum:	7.7	25	10	75	---	(see note 1)
Minimum:	6.3	---	---	---	120	---
TEST CONDITIONS:	7.0	25	---	---	120	---

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	6.6	7.5	A
© ---	Operation	2, 5	t = 300 seconds; ib = 10a; prr = 2,000 (min); tp = 5 µs (min)	0.65	II	---	---	---	---
© 1256	Electrode current (anode)	-	Eb = 175 Vdc	---	---	Ib	500	---	mAdc
---	Gas	3	Pp = 75 Vdc (min); t = 30 seconds	0.65	II	Δib	---	15	mAdc
1231	Peak emission	-	eb = 2,000 v	0.65	II	Is	16.0	---	a
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
1036	Bump	-	Hammer angle = 20°	---	---	---	---	---	---
---	Torque	4		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group C; operation; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Peak emission	-		---	---	Is	12.5	---	a
1331	Direct-interelectrode capacitance	6		---	---	Cpk	6.0	8.0	pF

NOTES:

1. Maximum forward pulse duration 5 µs; maximum inverse pulse duration 10 µs; pulse repetition rate 2,000 maximum. This tube is not recommended for power rectifier applications, but is intended primarily for clipper diode use in hard tube modulator circuits only where the inverse duration is not to exceed 10 µs.
2. The tube shall be operated in a pulsing circuit. The inverse pulse shall be of not less than 1 µs nor more than 5 µs duration. There shall be no evidence of sustained arc back or sputtering during this test.
3. The anode current shall not vary from the initial value at any time during the test by more than the specified amount.
4. The base shall be subjected to a gradually applied torque of 12.0-pound inches. This test shall not cause broken leads, broken welds, broken solder-joints, broken or cracked glass.
5. This test is to be the first test performed at the conclusion of the holding period.
6. © This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection (QCI), part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

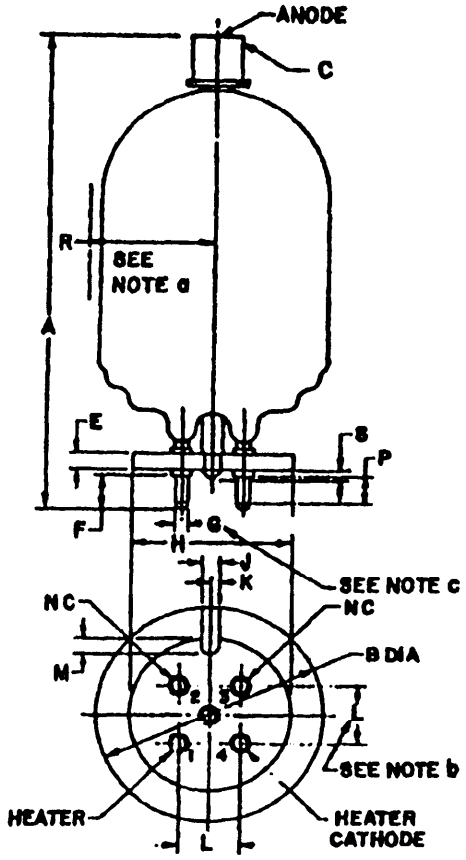
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2920)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.625 (142.88)	5.875 (149.23)
B		2.563 (65.10)
H	1.787 (45.39)	1.813 (46.05)
S		.125 (3.18)
© Quality conformance inspection, part 3 (see note d)		
C	Cap: C1-5 (EIA)	
G	.183 (4.65)	.191 (4.85)
J	.171 (4.34)	.203 (5.16)
K	.078 (1.98)	.110 (2.79)
M	.171 (4.34)	.203 (5.16)
P	.328 (8.33)	
R		1.406 (35.71)
Reference dimensions		
E	.188 (4.78)	
F	.406 (10.31)	
L	.688 (17.48)	

NOTES:
a. Eccentricity with respect to centerline of base.
b. The base shall be capable of entering a gage .25 (6.35 mm) thick having 4 holes of .214 (5.44 mm) dia located on .69 (17.53 mm) centers.
c. Solder should not extend laterally beyond cylindrical surface of studs.
d. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 719A.

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